

Current Implementation of Mortgage Rate Model

1) Raw Inputs

a) Historical 30Yr Primary Fixed-Rate

- i) The rates are published every Thursday and assumed to be backward looking. For example, the rates published on 9/18 (Thursday) should be the average rate from 9/11 (last Thursday) to 9/17 (Wednesday). Therefore, the rate published on 9/18, say 4.23%, is stored for 9/11 - 9/17.
- ii) Apply the Fee&Points adjustment on the mortgage rates. It should be $\text{RawRate} - (1 - \text{Fee\&Points})/4$. For example, the rate published on 9/18 is 4.23% with Fee&Points 0.5, then the mortgage rate used should be $4.23 - (1 - 0.5)/4 = 4.105\%$

b) Historical 30Yr Current Coupon Rate

The daily current coupon rate is calculated in the following fashion:

Suppose we want to calculate Fannie Mae current coupon. Suppose on 10/29/2014, we have the near-par, front-month TBA prices; FNMA2.5-TBA-NOV: 96.453125, and FNMA3-TBA-NOV: 100.023438, both settling on 11/13/2014. We also have corresponding back-month TBA prices; FNMA2.5-TBA-DEC: 96.21875, and FNMA3-TBA-DEC: 99.734375, both settling on 12/11/2014. The goal is to compute the price for a synthetic at-par TBA settling in 30 days with 25 days delay (specific to FNMAs), and then to use the price to derive the current coupon rate.

i) 30 Day Forward Prices

There are 15 calendar days between 10/29/2014 and 11/13/2014, 43 between 10/29/2014 and 12/11/2014. For FNMA2.5s, the forward price (30 days) is $(30-15)/(43-15) * (96.21875 - 96.453125) + 96.453125 = 96.32756696$, and for FNMA3s, is $(30-15)/(43-15) * (99.734375 - 96.21875) + 96.21875 = 99.86858282$.

ii) Payment Delay Adjustment

With 25-day delay adjustment, the prices are: $96.32756696 + (25-1)/360 * 2.5 = 96.49423363$ for FNMA2.5s, and $99.86858282 + (25-1)/360 * 2.5 = 100.0685828$ for FNMA3s.

iii) Monthly Par Rate

The implied yield from the prices is $(100 - 96.49423363) / (100.0685828 - 96.49423363) * (3.0 - 2.5) + 2.5 = 2.9904$.

iv) Yield Conversion

To convert the mortgage yield (monthly compounded) to bond equivalent yield (semi-annually compounded), we have $((1 + 2.9904/1200)^6 - 1) * 200 = \mathbf{3.009}$

	Price, Settle 1 = 11/13/2014	Price, Settle 2 = 12/11/2014	Step i: 30 Day Forward Prices	Step ii: Delay Adjusted Prices	Step iii: Monthly Par Rate	Step iv: Yield Conversion
FNCL 2.5 TBA	96.453125	96.21875	96.32756696	96.49423363	2.990406251	3.009098539
FNCL 3 TBA	100.023438	99.734375	99.86858282	100.0685828		

c) Historical Swap 10Yr

Daily data are obtained from yieldbook. a), b), and c) are used to calculate spreads.

d) Projected Swap 10Yr

We generate a vector of projected Swap 10Yr rates for each interest rate path from our interest rate model.

2) Spread Assumptions

$$CC\ Spread = Current\ Coupon - Swap\ 10Yr$$

$$Mortgage\ Spread = Mortgage\ Rate - Current\ Coupon$$

$$\widehat{M}_t = \widehat{C}_t + \alpha^t(C_0 - \widehat{C}_0) + \widehat{S}_t + \beta^t(S_0 - \widehat{S}_0) \quad (1)$$

- a) Currently in production we assume both spreads constant. The spreads are calculated using the most recent 2-week average. There is no "mark-to-market" mechanism. In terms of (1),

$$\widehat{C}_t = Swap10Yr_t + CC\ Spread$$

$$\widehat{S}_t = Mortgage\ Spread$$

$$\alpha = \beta = 0$$

- b) Alternatively, we also have facilities to switch to decaying spread, which is calculated as follows:

$$constSpread * e^{\left(-\frac{t}{6}\right)} + 0.01 * (1 - e^{\left(-\frac{t}{6}\right)})$$

where 0.01 is the assumed long term spread.

3) Data massaging

- a) Fill in mortgage rates for current week

As mentioned above, we only have mortgage rates data for previous week. For example, if the pricing date is 9/23/2014, the latest rate published is 9/18, which is used for 9/11 - 9/17, and we do not have "historical" rate for 9/18 - 9/23. We need to do the following:

- For the week 9/11 - 9/17, the spread = AvgMortgageRate[9/11 - 9/17] - AvgCurrentCoupon[9/11 - 9/17]. Store the weekly spreads somewhere.
- Do the same for the previous weeks and use the average of, say, previous 2 weeks, to avoid bumps. The "2" should be a parameter adjusted by the user.
- For the current week, 9/18 - 9/23, we have historical current coupon rates, so we use these along with the computed spread to get mortgage rates.

- b) Computing monthly average

With spread assumptions, in this example, we will have projected mortgage rates every month (9/24, 10/24, 11/24, ...) for 30 years. We need to do the following to interpolate them to get average monthly rates:

- For any month but current month (See Figure 1), the formula is simple. Suppose the available date (mortgage rate being MR_i) is the jth calendar day of that month, and there are x

calendar days in that month. Also suppose there are y calendar days between this available date and previous one (rate being MR_{i-1}), and z calendar days between this and the next (rate being MR_{i+1}). The formula is simply:

$$\text{AvgRate}_i = [(\text{MR}_y + \text{MR}_i)/2 * j + (\text{MR}_z + \text{MR}_i)/2 * (x - j + 1) - \text{MR}_i]/x$$

$$\text{where } \text{MR}_y = (\text{MR}_i - \text{MR}_{i-1}) * (y - j + 1)/y + \text{MR}_{i-1}$$

$$\text{and } \text{MR}_z = (\text{MR}_{i+1} - \text{MR}_i) * (x - j)/z + \text{MR}_i$$

ii. For the current month average rate, the calculation is more complicated because we need to use the daily rates for the dates before the pricing date in the same month. The rates calculated from 3.a. will be used and possibly some historical rates from 2 as well. Also note that here we should use business days instead to avoid errors.

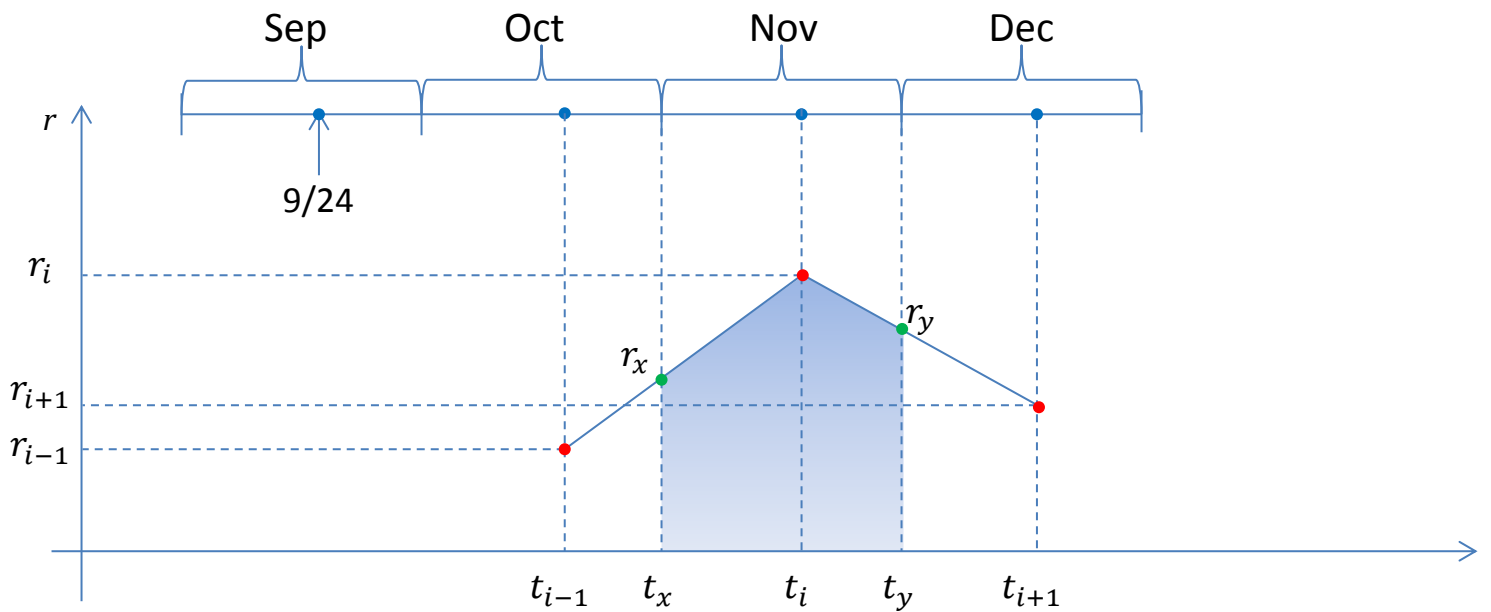


Figure 1